

Project II Rubric

E	S	R	N
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Rubric

Each item is marked ✓, ✓−, or ✗:

✓ = capable ✓− = mostly capable ✗ = not yet

General Rubric

Simulation Study Design (Part I)

Mark	Description
✓ ✓− ✗	(1) The base model and sampling procedures are correctly implemented and described with enough detail for replication.
✓ ✓− ✗	(2) The two chosen studies (A–D) are clearly identified and justified; the design is appropriate for the intended violation type and potential sources of bias are acknowledged.
✓ ✓− ✗	(3) Each violation scenario is accurately implemented in R with errors correctly generated.

Simulation Study Results (Part I)

Mark	Description
✓ ✓– ✗	(1) Coverage probabilities are reported in a well-organized table; bootstrap p-values for $H_0: \beta_1 = 0$ are compared to classical p-values; statistical tests are appropriate and multiple comparisons are interpreted carefully.
✓ ✓– ✗	(2) Average CI widths, bias of $\hat{\beta}_1$, and histograms of the sampling distribution of $\hat{\beta}_1$ are accurately reported; tables and figures are consistent with the narrative.
✓ ✓– ✗	(3) Results are interpreted with technically correct statistical reasoning; statistical and practical significance are distinguished.

Real Data Analysis (Part II)

Mark	Description
✓ ✓– ✗	(1) The dataset is clearly described and cited; the predictor is motivated through EDA; the research question is concise and the OLS design is appropriate.
✓ ✓– ✗	(2) Classical and bootstrap 95% CIs and p-values for $H_0: \beta_1 = 0$ are correctly computed and interpreted; agreement or disagreement between the two approaches is explicitly discussed.
✓ ✓– ✗	(3) Residual diagnostics are correctly produced and interpreted; OLS assumptions are assessed; the connection to the most analogous Part I scenario is argued with a well-reasoned conclusion about trust in the classical CI and p-value.

Validity of Conclusions (Part III & Conclusion)

Mark	Description
✓ ✓− ✗	(1) Conclusions follow logically from the results; interpretations are accurate and not overstated
✓ ✓− ✗	(2) Practical recommendations are grounded in simulation evidence

Report Quality, Replicability, and R Code

Mark	Description
✓ ✓− ✗	(1) Report is well-structured; all sources are cited and attributed; no issues of plagiarism or missing references.
✓ ✓− ✗	(2) Visuals are clear, labeled, and support the narrative; text is fully consistent with all tables and figures.
✓ ✓− ✗	(3) R script is clean, reproducible, and commented; <code>set.seed()</code> is used throughout; the analysis is replicable from the report alone.

Individual Rubric

Mark	Description
✓ ✓− ✗	(1) Completed assigned individual tasks thoroughly.
✓ ✓− ✗	(2) Actively participated in group collaboration and communication.

Grading Standards

- **E: Excellent:** Satisfactory with exceptional creativity or insight
- **S: Satisfactory:** Any combination of ✓ and ✓−
- **R: Revisions Needed:** Any ✗ on any criterion
- **N: Not Assessable:** Cannot grade due to too many missing components